Dataflow

Dataflow

- Google Dataflow is a fully managed, serverless service for building and executing data processing pipelines.
- It supports both batch and streaming data workflows.
  - Batch Mode: Processes finite, static datasets—for daily reports or file transformations.
  - Streaming Mode: Handles unbounded, continuously arriving data—for real-time analytics or alerts.
- Powered by the Apache Beam SDK, it offers a unified model for defining your data logic.

Serverless Infrastructure

- Dataflow automatically provisions and tears down worker VMs as needed. You don't manage infrastructure.
- Billing is based only on the resources consumed during pipeline execution.
- Optimizes resource use by dynamically scaling worker VMs in response to workload.
- Supports dynamic work rebalancing to improve pipeline efficiency.
- Dataflow uses the open-source Apache Beam SDK for pipeline definitions.

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Scalability & Integration

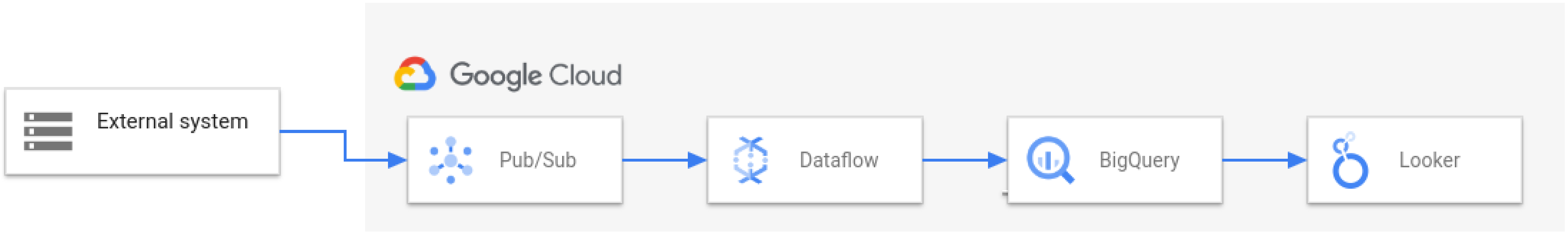
- Capable of supporting pipelines with thousands of worker VMs and petabytes of data.
- Designed to handle enterprise-scale workloads efficiently.
- Dataflow templates are saved pipeline definitions for reuse across jobs.
- Templates support parameterization (e.g., dynamic input/output settings).
- Seamlessly connects with services like:
    - BigQuery for analytics.
    - Cloud Storage for ingest/output.
    - Cloud Pub/Sub for streaming inputs.
    - Other systems via Beam IO connectors (e.g., Datastore, Spanner).

Dataflow pipeline

- A pipeline consists of:
  - **Sources:** where data is read from (e.g., Cloud Storage, Pub/Sub, BigQuery).
  - **Transforms:** operations like filter, map, group.
  - **Sinks:** where output is written (e.g., BigQuery, Cloud Storage).
- Data is passed through PCollections, which represent datasets within the pipeline.
- Offers **exactly-once** processing semantics by default.

ETL and BI solution using Dataflow

- Pub/Sub ingests data from an external system.
- Dataflow reads the data from Pub/Sub and writes it to BigQuery. During this stage, Dataflow might transform or aggregate the data.
- BigQuery acts as a data warehouse, allowing data analysts to run ad hoc queries on the data.
- Looker provides real-time BI insights from the data stored in BigQuery.
- For basic data movement scenarios, you might run a Google-provided template. Some templates support user-defined functions (UDFs) written in JavaScript. UDFs let you add custom processing logic to a template.



Use Case

- Detect suspicious credit card transactions as soon as they happen so action can be taken immediately.
- Data Ingestion, every transaction is sent to Pub/Sub in real time.
- Dataflow pipeline reads each transaction as it arrives, cleans & prepares the data (e.g., amount, location, device info), runs it through a machine learning model (e.g., fraud detection).
- If transaction looks risky → Send to Alerts topic (for banks, apps), store all results in BigQuery for dashboards & reports.
- Dataflow automatically scales with traffic and gives real-time job monitoring.

Autoscaling

- Capable of supporting pipelines with thousands of worker VMs and petabytes of data.
- Designed to handle enterprise-scale workloads efficiently.
- Dataflow optimizes pipeline execution through automatic scaling and load balancing.
- Autoscaling helps optimize cost by right-sizing resource usage.
- Consumption-based billing—charges for vCPU, memory, shuffle, and network usage.